

Initialize the environment

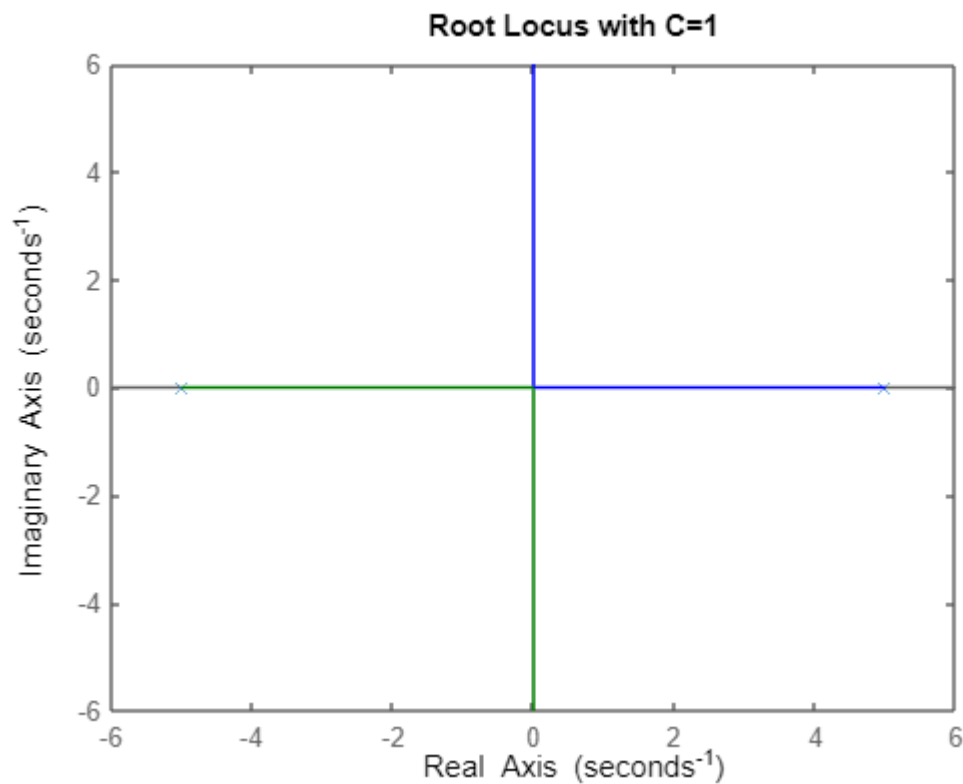
```
clear all;  
close all;
```

Define the Laplace variable and the system

```
s = tf('s');  
omega = 5;  
G = omega^2/(s^2 - omega^2);
```

Plot root locus with C=1

```
C = 1;  
figure;  
rlocus(C*G);  
title('Root Locus with C=1');
```



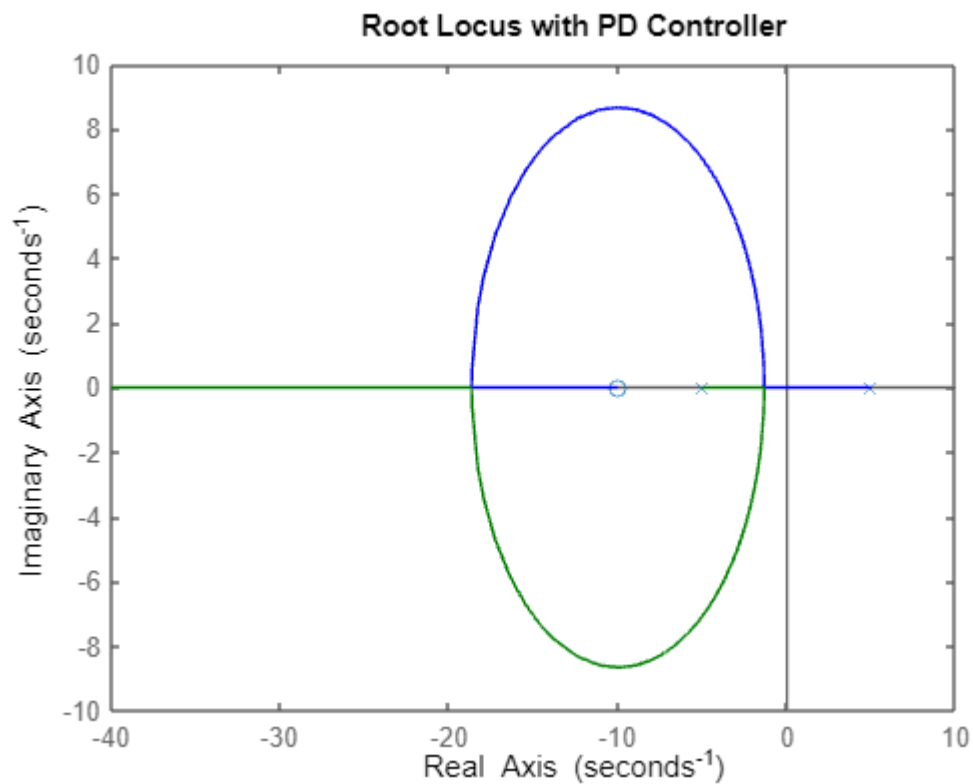
Design with a PD controller

```
kp = 10;  
kd = 1;  
C = kp + kd*s;
```

Plot root locus with PD controller

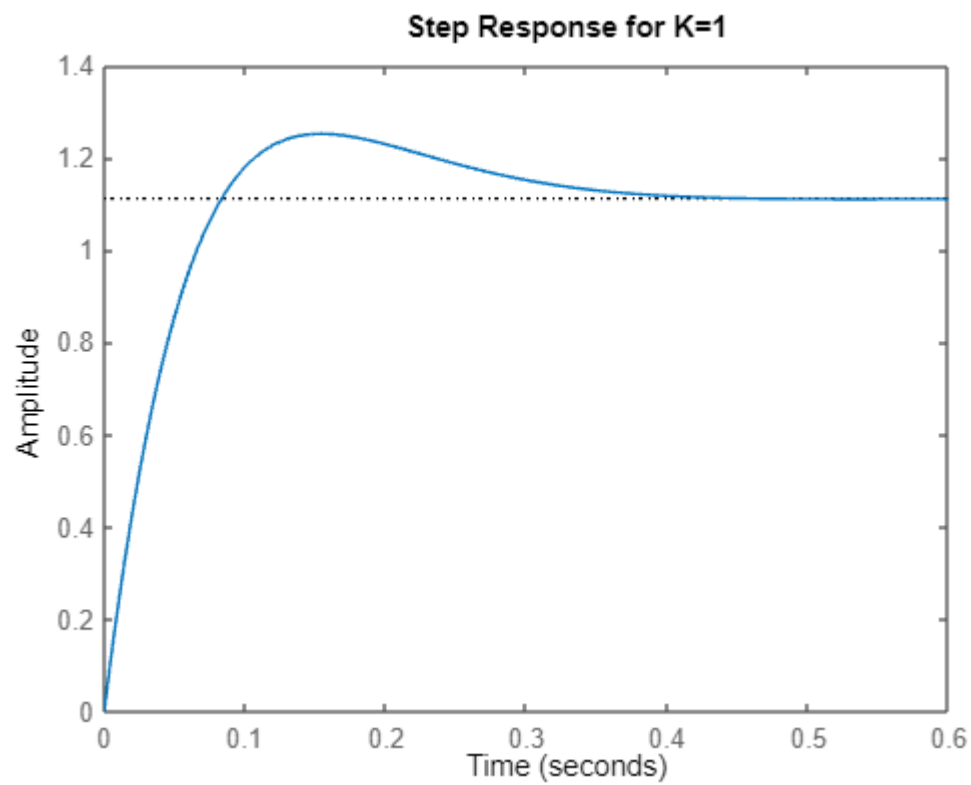
```
figure;
```

```
rlocus(C*G);  
title('Root Locus with PD Controller');
```



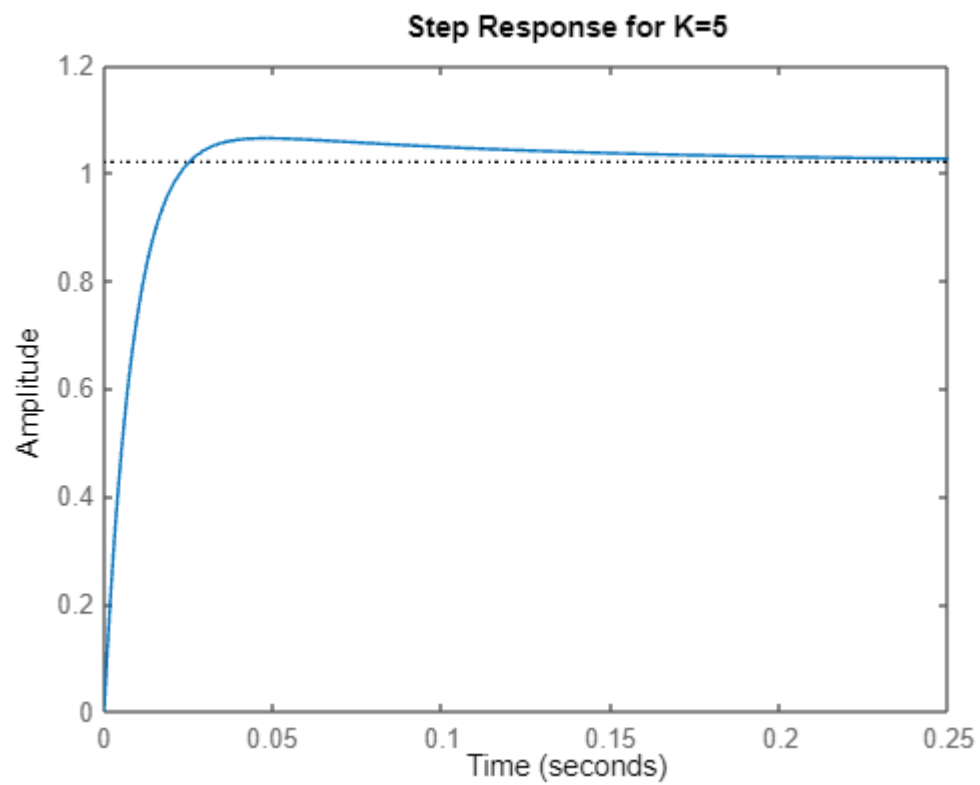
Evaluate step response for K=1

```
K = 1;  
C = K*(kp + kd*s);  
GCL = feedback(C*G, 1);  
figure;  
step(GCL);  
title('Step Response for K=1');
```



Evaluate step response for K=5

```
K = 5;  
C = K*(kp + kd*s);  
GCL = feedback(C*G, 1);  
figure;  
step(GCL);  
title('Step Response for K=5');
```

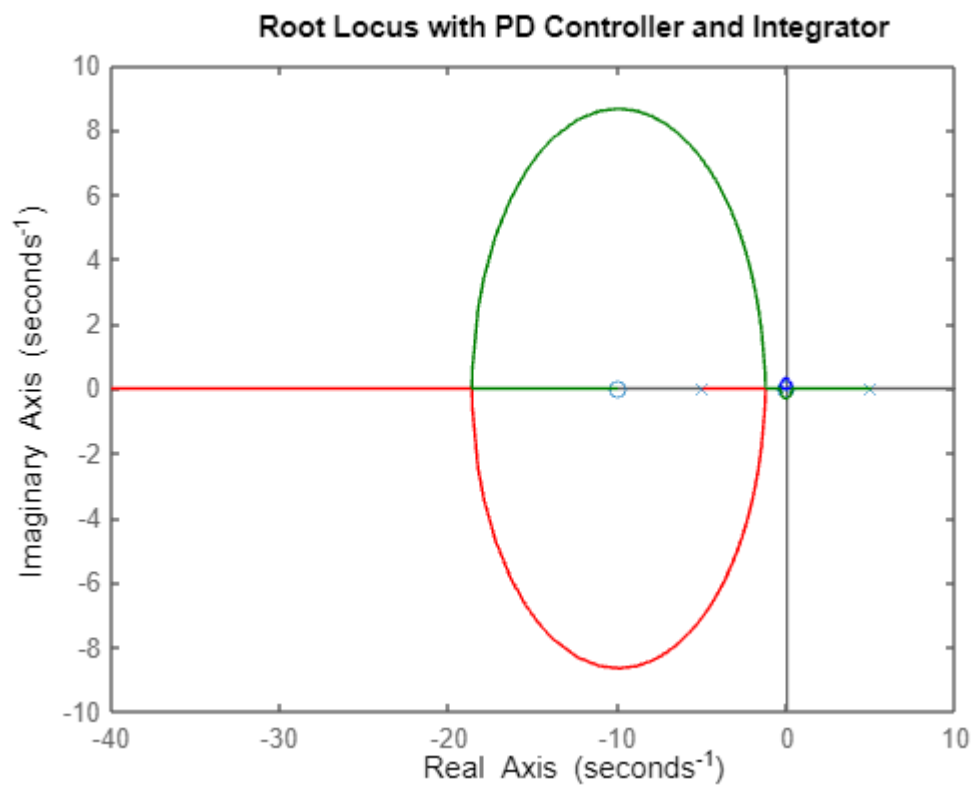


Design with PID controller (the I is for zero steady-state error)

```
ki = 0.1;  
C = kp + kd*s + ki/s;
```

Plot root locus with PID controller

```
figure;  
rlocus(C*G);  
title('Root Locus with PD Controller and Integrator');
```



Evaluate step response with designed K

```
K = 0.13;
C = K*(kp + kd*s + ki/s);
GCL = minreal(feedback(C*G, 1));
figure;
step(GCL);
title('Step Response with PD Controller, Integrator, and K=0.13');
```

